TATHYA DUBEY

Software Development Engineer (Full-Stack & Distributed Systems) | B.Tech CSE, 2027
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SUMMARY

CS undergraduate (CGPA 8.71) focused on backend and systems engineering: protocol-level networking, distributed queues and caching, type-safe full-stack applications, and compiler internals. Comfortable owning a system from schema design through deployment. Seeking SDE / Full-Stack roles.

TECHNICAL SKILLS

Languages & Core: TypeScript, Python, JavaScript, C, C++, SQL, POSIX, Recursive-Descent Parsing, ASTs.
Backend & Distributed Systems: Node.js, FastAPI, Next.js, React, PostgreSQL, Supabase, Redis (Lua/Pub-Sub), Celery, Kafka/RabbitMQ, Native SQLite, Docker, REST, WebSockets, SSE, Alembic, Zustand, Zod.
AI & Data Layer: LangGraph Multi-Agent Orchestration, pgvector, Hybrid RAG & Cross-Encoder Reranking, Semantic Caching, LLM Gateways, OpenTelemetry.

EXPERIENCE

THEKULTHOUSE | Co-Founder & Technical Lead **Jun 2026 – Present**
- Co-founded a digital engineering agency delivering production B2B SaaS platforms, custom booking engines, and full-stack web infrastructure for commercial clients.
- Owned the full lifecycle: Postgres/Supabase schema design, RBAC security, CI/CD, Docker VPS deployment.

HostingSpell | Cybersecurity Intern **Nov 2025 – Dec 2025**
- Audited web applications for injection flaws, broken authentication, and security misconfigurations; implemented hardened patches and strict CSP/CORS policies.

PROJECTS

Pulse | Zero-Dependency Uptime & Incident Monitor | *Node.js 22, node:sqlite, Vanilla JS, SSE*
- Built a self-hosted uptime monitor with zero npm packages, using only Node.js built-ins and node:sqlite.
- Wrote custom HTTP(S), TCP, DNS, TLS-expiry, and ICMP probes, plus a dead-man's-switch heartbeat API.
- Designed a state-machine incident engine with failure thresholds, maintenance windows, HMAC-signed webhook/SMTP alerts with retries, and live p50/p95/p99 latency over SSE.

Drewit | Collaborative Infinite Canvas Workspace | *TypeScript, Next.js, React, Canvas API*
- Built an infinite collaborative whiteboard with embedded Markdown, runnable multi-language code, and spreadsheet blocks with formula parsing (=SUM, =AVG, =IF) and CSV I/O.
- Added an Explore feed, Communities, and bookmarking, plus a browser-local file drawer that keeps private PDFs/DOCX away from board collaborators.

DevPulse & SMTEKE | AI Code Auditor & Multi-Tenant RAG | *FastAPI, LangGraph, Redis, pgvector*
- Hosted a LangGraph multi-agent reviewer on FastAPI/Celery/Docker that clones GitHub repos, retrieves security rules via pgvector RAG, and proposes line-level fixes using Anthropic, Gemini, and Ollama.
- Fed a multi-tenant S3/Kafka ingestion pipeline into a query router with Redis semantic caching, BM25 + vector hybrid search, and cross-encoder reranking.

B2B Factory ERP | 10-Module Manufacturing Platform | *Next.js, Supabase, Three.js, Zustand, Zod*
- Built inventory, production, and granular RBAC modules on Supabase/PostgreSQL with end-to-end Zod validation, Zustand cross-module state, and Three.js 3D factory-floor views.

EPL (English Programming Language) | In Progress | *C, POSIX, CLI, VS Code Extension*
- Wrote a natural-language programming language in C: handwritten lexer, recursive-descent parser, AST evaluator, declarative UI engine (.epf), and a VS Code extension.

EDUCATION & CERTIFICATIONS

Indus Institute of Technology and Engineering **Expected 2027**
B.Tech in Computer Science & Engineering | **CGPA: 8.71 / 10.0**
- AWS Academy Graduate: Machine Learning Foundations (2026) – credly.com/go/V436gx4Q
- AWS Academy Graduate: ML for Natural Language Processing (2026) – credly.com/go/l6gon3el